Aeronautics and Space Technology Club

IIT (ISM) Dhanbad

Signal Processing Ladder: A Hands-On Problem Series from Sensors to Machine Learning

Problem Statement 01 · Part A: Foundations Level 1

Track: Signal Measures, Sensor Models, Noise Statistics, Allan Deviation

Prerequisites: basic Python, high-school calculus · **Estimated effort:** 8–12 hours

Signals, Sensors and Noise: Characterise the Raw Data Before You Process It

Every flight computer, ground station and telescope starts with a sensor that turns a physical quantity into a noisy voltage. Build the vocabulary (RMS, power, decibels, SNR), model a real sensor with sensitivity, offset and noise, learn when averaging helps and when it stops helping, and measure a gyroscope's noise with the Allan deviation.

General Instructions

- **Thorough Review:** Read the whole problem statement, including the theory, before writing any code. Do the short analytic derivations *by hand first*; the code is there to confirm and extend your understanding, not to replace it.
- **Pacing:** This is a self-paced problem. The split below is a guide for the effort estimate in the header.

Stage	Phase	What you should have at the end of it
1	Think (∼20%)	Theory understood; every analytic result derived; signals, units and sampling rates written down.
2	Build (∼30%)	Core code working end to end on the baseline data and validated against the analytic results.
3	Experiment (∼30%)	Full sweeps, stress tests and the real or synthetic datasets processed; first plots and numbers.
4	Refine (∼20%)	Weaknesses fixed, honest analysis and memo written, repository folder clean.

- **Compute and Tools:** Everything is designed to run on a laptop CPU in Python 3.10+ (NumPy, SciPy, Matplotlib, and where stated pandas, Astropy, scikit-learn, PyWavelets and PyTorch). No GPU and no hardware are required; where a problem offers an optional hardware experiment it is marked as a bonus. A full run should never need more than a few minutes; if yours does, vectorise before you scale up.
- **Data:** Each problem lists (i) a *synthetic generator* from `sp_common.synth` (offline, seeded, with known ground truth) and (ii) *real datasets* to download yourself. Develop and validate on the synthetic data first, then repeat the key result on real data. Never commit large raw data files to the repository.
- **Submission Format:** Push your work to `solutions/01_signals_sensors_and_noise/` in the repository: (i) clean, commented, reproducible Jupyter notebook(s) with fixed random seeds, (ii) a short written report (`report.pdf` or `report.md`) containing the requested analysis sections, and (iii) all figures in a `figures/` sub-folder. The report must be understandable without running the code.

- **Originality:** All code, analysis and written explanations must be your own work. Any external library, snippet, paper or AI tool you use must be cited in the report. You should be able to explain every line you submit.
- **Honesty over Hype:** Validate against analytic results *before* trusting a number. A modest or negative result that is correctly analysed is worth more than an inflated claim. Report error bars, false-alarm rates and failure cases, not only your best run.
- **Units:** Write the units of every quantity and state in the first notebook cell whether a frequency is in Hz or rad/s and whether it is normalised to the sampling rate. Most mistakes in signal processing are factors of 2π, factors of 2 in one-sided spectra, or an off-by-one in the frequency axis.
- **Getting Stuck:** Re-read the theory section, check the self-check targets, then ask your mentor or open an issue in the repository. Describe what you expected, what you got, and what you already tried.

Background and Motivation

A spacecraft never measures "angular rate" or "temperature". It measures a voltage, a count or a photon arrival time, and everything we call data is the output of a *sensor model*: a physical quantity multiplied by a sensitivity, shifted by an offset, and corrupted by noise that has several different origins. Before any filter, transform or neural network is applied, an engineer must be able to answer three questions about the raw stream: how large is the signal, how large is the noise, and what kind of noise is it (white, drifting, impulsive)? The answers decide the sampling rate, the amplifier gain, the averaging time and the choice of filter.

This problem builds that vocabulary from the ground up using nothing more than NumPy. You will work with a vibration-like sinusoid in noise, a calibrated sensor, and a rate gyroscope whose noise is dominated by different mechanisms on different time scales. The Allan deviation, which you implement from its definition, is the standard tool of inertial navigation for telling these mechanisms apart, and it reappears in problem 20 where it sets the process noise of a Kalman filter.

Theoretical Framework

1. Signal Classes and Measures

A continuous-time signal $x(t)$ is defined for all real t; a discrete-time signal $x[n] = x(nT_s)$ only at integers n, with sampling period $T_s = 1/f_s$. For a record of N samples,

$$\bar{x} = \frac{1}{N}\sum_{n=0}^{N-1} x[n], \qquad E_x = \sum_{n=0}^{N-1} x[n]^2, \qquad P_x = \frac{E_x}{N}, \qquad x_{\mathrm{rms}} = \sqrt{P_x}. \tag{1}$$

For a sinusoid $A\sin(2\pi f_0 t + \phi)$ the mean power is $A^2/2$ and $x_{\mathrm{rms}} = A/\sqrt{2}$; for a symmetric square wave $x_{\mathrm{rms}} = A$ and for a triangle wave $x_{\mathrm{rms}} = A/\sqrt{3}$ (derive all three). The *crest factor* $x_{\mathrm{peak}}/x_{\mathrm{rms}}$ tells you how much headroom an amplifier or ADC needs.

2. Decibels and Signal-to-Noise Ratio

Decibels compare *powers*; amplitudes are squared first:

$$G_{\mathrm{dB}} = 10\log_{10}\frac{P_2}{P_1} = 20\log_{10}\frac{A_2}{A_1}, \qquad \mathrm{SNR}_{\mathrm{dB}} = 10\log_{10}\frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}}. \tag{2}$$

Doubling an amplitude adds 6.02 dB; doubling a power adds 3.01 dB. Absolute levels use a reference, for example $\mathrm{dBm} = 10\log_{10}(P/1\,\mathrm{mW})$.

3. A Sensor Model

A linear sensor with a quantity x at its input produces

$$y[n] = S\,x[n] + b + \eta[n], \qquad \eta[n] \sim \mathcal{N}(0, \sigma^2), \tag{3}$$

with sensitivity S (for example V/g), offset b and white noise of standard deviation σ. Given calibration points (x_k, y_k) taken with known inputs, ordinary least squares gives $\hat{S} = \sum(x_k - \bar{x})(y_k - \bar{y}) / \sum(x_k - \bar{x})^2$ and $\hat{b} = \bar{y} - \hat{S}\bar{x}$ with $\mathrm{Var}(\hat{S}) = \sigma^2 / \sum(x_k - \bar{x})^2$. Noise is quoted either as an RMS value over a stated bandwidth or as a *noise density* n_d in $\sqrt{\text{Hz}}$; for white noise sampled at f_s with per-sample standard deviation σ, the one-sided density is $n_d = \sigma\sqrt{2/f_s}$ and conversely $\sigma = n_d\sqrt{f_{\mathrm{BW}}}$ with $f_{\mathrm{BW}} = f_s/2$.

4. Averaging, Robust Statistics and Their Limits

Averaging M independent samples reduces the noise standard deviation to $\sigma/\sqrt{M}$, i.e. it gains $10\log_{10} M$ decibels of SNR. This stops when the noise is not independent: a slow drift (flicker or random-walk noise) does not average down, so an optimum averaging time exists. Impulsive outliers (spikes from a particle hit or a loose connector) wreck the sample standard deviation; the *median absolute deviation* $\mathrm{MAD} = \mathrm{median}\,|x_i - \mathrm{median}(x)|$ gives the robust estimate $\hat{\sigma} = 1.4826\,\mathrm{MAD}$ for Gaussian data.

5. Allan Variance

Split a record into consecutive clusters of duration τ, let $\bar{y}_k$ be the mean of cluster k, and define

$$\sigma_A^2(\tau) = \frac{1}{2}\left\langle (\bar{y}_{k+1} - \bar{y}_k)^2 \right\rangle_k . \tag{4}$$

On a log–log plot the slope identifies the noise type: white rate noise (angle random walk, ARW) gives $\sigma_A(\tau) = \mathrm{ARW}/\sqrt{\tau}$ (slope $-\frac{1}{2}$), flicker bias noise gives a flat floor (the *bias instability*), and rate random walk gives slope $+\frac{1}{2}$. For per-sample white noise of standard deviation σ at rate f_s, $\mathrm{ARW} = \sigma/\sqrt{f_s}$ in $^\circ/\sqrt{\text{s}}$ if σ is in $^\circ$/s.

Data and Simulation Specification

Synthetic data	`sp_common.synth.sensor_signal(fs, duration, f0, amp, noise_rms, drift_per_s, spike_rate)` for Tasks 1–4; `sp_common.synth.gyro_noise(fs=100, duration=3600, sigma_white=0.02, sigma_rrw=1e-4, sigma_bias=0.005, tau_bias=200)` for Task 5 (units °/s; the ground-truth components are returned separately).
Reference signal	$f_0 = 5\,\text{Hz}$, $A = 1$, $f_s = 1000\,\text{Hz}$, $\sigma = 0.2$ (so SNR $\approx 11\,\text{dB}$).
Calibration	Known inputs $x \in \{-2, -1, 0, 1, 2\}\,g$, true $S = 0.3\,\text{V}/g$, $b = 12\,\text{mV}$, $\sigma = 5\,\text{mV}$, ten readings per input.
Real data (choose one)	(i) Record your own: install the free *Phyphox* app on a smartphone, place it flat and still on a table, and log the accelerometer for 10 minutes (export CSV). (ii) The EuRoC MAV visual-inertial dataset (ETH Zurich, ASL): use the stationary start of one IMU recording at 200 Hz.
Real-data checks	Compare the noise density you measure with a typical MEMS accelerometer datasheet value (order of 100–400 $\mu g/\sqrt{\text{Hz}}$).

Your Task

1. **Signal measures (Stage 2).** Implement Eq. (1), crest factor and decibel functions from scratch. Verify numerically $x_{\text{rms}} = A/\sqrt{2}$, A, $A/\sqrt{3}$ for sine, square and triangle waves and report the crest factors. Show that adding two uncorrelated signals adds their *powers*, not their amplitudes.
2. **Sensor calibration.** Simulate the calibration experiment, fit $\hat{S}$ and $\hat{b}$ by least squares and compare $\sqrt{\text{Var}(\hat{S})}$ from theory with the spread over 2000 repeated experiments. Explain which input values would make $\hat{S}$ most precise for a limited number of readings.
3. **Noise statistics.** Histogram the noise of the reference signal, overlay the Gaussian density and make a quantile–quantile plot. Add random spikes (`spike_rate`$= 2/\text{s}$) and compare the sample standard deviation with 1.4826 MAD; explain the difference.
4. **Averaging and its limit.** Verify $\sigma/\sqrt{M}$ by averaging non-overlapping blocks of length $M = 1, \ldots, 10^4$ of pure white noise (log–log plot). Repeat with a linear drift of 0.05/s added and find the block length at which averaging stops helping.
5. **Allan deviation of a gyroscope.** Implement Eq. (4) (non-overlapping and overlapping versions) and apply it to the synthetic gyro for τ from 0.01 s to 1000 s. Read off ARW, the bias-instability floor and the random-walk branch, and compare with the generator's true parameters.
6. **Real sensor.** Process your smartphone or EuRoC data: histogram, noise density in $\text{m/s}^2/\sqrt{\text{Hz}}$, Allan deviation. Is the noise white? What does the Allan curve say about the best averaging time?
7. **Dynamic range.** For an accelerometer with range $\pm 16\,g$ and a 16-bit ADC compute the LSB, the quantisation-limited dynamic range in decibels, and compare the LSB with your measured noise: is the sensor noise-limited or quantisation-limited?

Self-check targets

A unit sinusoid has $x_{\text{rms}} = 0.7071$ and crest factor 1.414; the reference signal has SNR ≈ 10.97 dB. Averaging $M = 100$ white samples reduces σ by a factor 10 (20 dB of SNR). For Gaussian data $1.4826\,\text{MAD} \to \sigma$. The synthetic gyro has ARW $= \sigma_{\text{white}}/\sqrt{f_s} = 0.002\,^\circ/\sqrt{\text{s}}$ (that is $0.12\,^\circ/\sqrt{\text{h}}$), so $\sigma_A(1\,\text{s}) \approx 0.002\,^\circ/\text{s}$ with slope $-\frac{1}{2}$ at short τ.

Aerospace Application Link

A small satellite that loses its star tracker must coast on its gyroscope. The pointing error from white rate noise grows as $\sigma_\theta(T) = \text{ARW}\sqrt{T}$, but an uncorrected constant bias b grows as bT. Using your Allan analysis, compute for $T = 1$ min, 10 min, 1 h the angle error from the ARW alone and from the measured bias-instability floor, and state for how long the satellite can stay within a pointing budget of 1° without an external reference. This number sets how often the attitude filter of problem 20 must be corrected.

Deliverables

D1 **Measures notebook:** signal measures, crest factors and decibel conversions with the three waveform checks.

D2 **Calibration and noise statistics:** least-squares calibration with theoretical versus empirical uncertainty, histogram and Q–Q plots, standard deviation versus MAD with spikes.

D3 **Averaging study:** $\sigma(M)$ log–log plot with and without drift and the optimal averaging length.

D4 **Allan analysis:** non-overlapping and overlapping Allan deviations of the synthetic gyro with ARW, bias instability and random-walk parameters extracted and compared with the truth.

D5 **Real sensor report:** noise density and Allan deviation of your own or EuRoC data, and the dynamic-range calculation.

D6 **Memo (250 words):** "How long may the satellite coast on its gyroscope?" Use your pointing-error numbers and cite at least two external sources.

Bonus (a) Add confidence intervals to the Allan deviation ($\sigma_A/\sqrt{2(N_\tau - 1)}$ as a first estimate, then a bootstrap). (b) Show that the Allan deviation of a sine-wave vibration produces a characteristic bump and explain it. (c) Hardware: log an MPU-6050 or similar IMU with a microcontroller and repeat the analysis in a thermally stable box and next to a warm object.

Learning Resources

Signals and Measurement

- Oppenheim, A.V., Willsky, A.S. *Signals and Systems*, Ch. 1 (signal classes, energy and power).
- Smith, S.W. *The Scientist and Engineer's Guide to Digital Signal Processing*, Ch. 2 (statistics, noise and signal-to-noise ratio); free at `https://www.dspguide.com`.
- Horowitz, P., Hill, W. *The Art of Electronics*, chapter on low-noise techniques (noise sources and noise density).

Allan Variance and Inertial Sensors

- Allan, D.W. (1966). "Statistics of atomic frequency standards." Proc. IEEE 54, 221.
- IEEE Std 952-1997, *Specification Format Guide and Test Procedure for Single-Axis Interferometric Fiber Optic Gyros* (Annex on Allan variance and noise terms).
- El-Sheimy, N., Hou, H., Niu, X. (2008). "Analysis and modeling of inertial sensors using Allan variance." IEEE Trans. Instrum. Meas. 57, 140.

Data and Tools

- Phyphox (smartphone sensor logging): `https://phyphox.org`
- EuRoC MAV dataset: `https://projects.asl.ethz.ch/datasets/doku.php?id=kmavvisualinertialdata`
- NumPy, Matplotlib: `https://numpy.org/doc/`, `https://matplotlib.org/stable/`; AllanTools (for cross-checking only): `https://allantools.readthedocs.io`

Evaluation

Score your own submission out of 100 on five equally weighted criteria, then ask your mentor to score it independently and compare.

Criterion	What to look for	Marks
Conceptual depth	Correct mathematics and physics; every result explained through a mechanism (why the spectrum looks like this), not only through code behaviour.	/20
Implementation quality	Correct, modular, commented, reproducible code; no unexplained magic numbers; analytic checks and self-check targets pass.	/20
Aerospace-application integration	The signal-processing result is carried through to a measurable consequence for a sensor, a link, a flight system or a detection decision, as the task requires.	/20
Analysis and critical thinking	Error bars, convergence, comparison to theory, honest treatment of limitations, false alarms and failure cases.	/20
Communication	Clear report, labelled plots (units, legends, captions) and a memo that a non-expert student could follow.	/20

Bonus tasks are not part of the 100 marks. A bonus attempted badly does not cost you marks, and a well-executed one is the best sign that you own the material.